

DXGAT / RampVision — technical audit: the current state

6 October 2026 · Valentyn Fedorov · what the system runs today: the modules and what each checks, the models behind them and how old they are, when each check is active during a turnaround, how long everything takes, and where the same logic is written several times. Based on the architecture review of 7 September and the measurements of 8 September – 1 October.

1 · What runs today and how long it takes

Everything is post-processing. Two cameras at the gate record 1080p at 8 fps in one-minute chunks and upload them; after the turnaround the chunks are merged into one full-turn video per camera; then three jobs run one after another on that file — the General Model (detects everything on the apron), the trackers (follow the aircraft, belt loaders, GSE and people), and one job per check — and the verdicts (Pass / Fail / Not observed) land in MongoDB and on RampVision. **A result arrives about 24 hours after the recording:** on the median of 148 production videos, 9.5 h for the last chunk to reach the bucket, 2.5 h of merge, 2.4 h of General Model, 1.3 h of trackers and 0.9 h of modules. Each GPU job runs on one Tesla T4 with about 4 CPU cores and 12 GB of memory.

stage	what it does	models	time on a typical (87-minute) video
merge	finds, orders and merges the chunks into one video	a small classifier for the capture boundary	1.7 h waiting + 0.8 h running
General Model	detects aircraft, parts, equipment, cones, chocks, people on every frame; decides the camera, the aircraft type and the airline; reads the video twice	3 YOLO detectors, MobileSAM, an airline detector, two small classifiers	2.4 h (207 ms per frame — 1.7× slower than the recording)
trackers	follows every object over time: identity, stops and movement, who is at which door	2 re-identification networks, 2 segmentation networks, optical flow	1.3 h (112 ms per frame)
modules (27)	the check logic, one job per module; 13 of them also look at the pixels	13 of 27 carry models of their own (pose, classifiers, segmentation)	0.9 h in total; 7 s to 6 min per module on the reference event
reporting	writes verdicts, reports and the smart timeline	—	minutes

2 · The shared models (run once per video for all checks)

model	what it is for	version	when it runs	size MB	time per frame
GM YOLOv8m (apron classes)	the main detector: aircraft and parts, cones, chocks, equipment, people	March 2024	every frame	6.2	8.5 ms
Chocks YOLOv8 v4.3	a second detector just for chocks, at higher resolution	v4.3	every frame	22.6	3.0 ms
Vehicle YOLOv8m	a third detector for vehicles and ground equipment	September 2023	every frame	52.0	5.8 ms
MobileSAM vit_t (main-aircraft mask)	an outline of the main aircraft for its movement state	—	every frame with the main aircraft	40.7	small
Entity (airline) YOLOv5	which airline (six logos)	—	every 8th frame, up to 40 hits	14.7	small
Camera cone/wing EfficientNet-B0 v1.8.1	which camera this is (cone or wing)	v1.8.1	every post-arrival frame with the main aircraft, CPU	16.3	small
Tail classifier	aircraft type (jet / aircraft)	—	aircraft type vote	0.4	small

Trackers: two re-identification networks (one trained for people; for belt loaders and GSE a generic ImageNet ResNet-34 used as is), two segmentation networks (aircraft; belt loader and GSE), classic optical flow on up to 250 points per object, and a noise estimate of the frame every two seconds. Together 26 ms per frame on a modern desktop GPU, 112 ms on the production T4.

3 · The checks: what each module verifies, when it is active, what it looks at

27 module repositories implement the client's 30 checks (the chock module gives three verdicts; “proper belt loader approach” is derived from two others). Window: the part of the turnaround the check is active in. Looks at: the detections it reads from the General Model and the objects it follows through the trackers. Pixels: whether the module also reads the images itself.

M	module	the check	window · how it decides	looks at (General Model)	follows (trackers)	pixels	own models
M01	3-stop brake check	3 stop brake check	belt loader at the door · verdict at the event	GSE, pushback, trailer	aircraft, belt loader	no	none
M02A	aircraft chocks (3 verdicts)	Nose gear chocks applied immediately; Main gear chocks re...	before departure · verdict at the event	pushback, chocks, cones, trailer, fuel truck, ladder, front whe...	aircraft, people, GSE, belt loader	yes	frame-difference chock classifier (...)
M03	cargo doors opened	All cargo bin doors opened and verified	belt loader at the door · Fail if not seen in time	front door, rear door, GSE, pushback, trailer, fuel truck, vehi...	aircraft	no	none
M04	belt loader forward chock	Belt loader forward chock remained in place until unit is...	belt loader at the door · verdict at the event	chocks, cones, rear wheel, engine, rear door, wing, pushback, t...	aircraft, belt loader	no	none
M05	belt loader rear cone	Beltloader rear cone positioned after BL is in place	belt loader at the door · verdict at the event	BL cone, engine cone, wing cone, cones, rear wheel, engine, rea...	aircraft, belt loader	no	none
M06	chocks and cones staged	Chocks and cones available and staged for arrival	before arrival · judges the window before the moment	obstacles, side obstacles, belt loader, GSE, pushback, cones, c...	aircraft	no	none
M07	conditioned air removed	Conditioned air removed 10 mins prior to departure and pr...	before departure · judges the window before the moment	trailer, GSE, belt loader, front wheel, nose, pushback	aircraft	yes	AC hose detector (YOLO11s-seg, 1 cl...
M08	cones removed after GSE clear	Cones removed only after all GSE is clear and chocked	before departure · verdict at the event	engine cone, wing cone, tail cone, GSE, pushback, trailer, belt...	aircraft, belt loader	no	none
M09	cones placed	Cones placed in proper positions and timely	arrival · Fail if not seen in time	wing cone, engine cone, tail cone, tail, GSE, pushback, trailer...	aircraft	no	none
M10	crew present	Crew present 10 minutes prior to aircraft arrival	before arrival · judges the window before the moment	people, GSE, pushback, trailer	aircraft	no	none
M11	FOD walk	FOD walk completed	before arrival · judges the window before the moment	front wheel, GSE, belt loader, pushback, tail, nose	aircraft, people	yes	MoveNet SinglePose Thunder v4
M12	GSE parked and chocked	Motorized GSE parked and properly chocked	belt loader at the door · verdict at the event	chocks, pushback, trailer, fuel truck, cones, ladder	GSE, people, belt loader	yes	frame-difference chock classifier; RTMPose-l body pose (17 keypoints)
M13	hand signals	All GSE guided into aircraft using approved hand signals	arrival · verdict at the event	front door, rear door, pushback, rear wheel, engine, wing, trai...	aircraft, belt loader, people	yes	HRNet-W48 top-down heatmap pose (CO...
M14	handrails on GSE used	Handrails on GSE being used	belt loader at the door · verdict at the event	people, trailer, GSE, engine, rear wheel, rear door	aircraft, people, belt loader	yes	EfficientNetB0 handrail-extension c...; SegFormer-style handrail segmentati...; DWPose-l whole-body pose (133 keypo...; Hand-contact classifier (blind chec...
M15	lead marshaller and wing walkers	Lead marshaller and wing walkers in correct position	before arrival · judges the window before the moment	tail, wing, belt loader, GSE, pushback, trailer, obstacles	aircraft, people	no	HRNet-W48 top-down heatmap pose (CO...; worker role classifier (SVC); walk-along classifier (RandomForest); walker movement-start models (left,...
M16	pushback pin verification	Pushback operator verifies steering bypass pin installati...	before departure · verdict at the event	pushback, nose, front wheel, belt loader, GSE, trailer, obstacl...	aircraft, people	yes	SimCC ResNet-50 top-down pose (COCO...
M17	post-arrival walk-around	Post-arrival walk around inspection completed accurately;...	arrival · needs the whole event	belt loader, fuel truck, GSE, ladder, trailer, rear wheel, tail...	aircraft, people, belt loader	yes	YOLOv8m-pose (worker keypoints; sin...; Depth Anything V2 metric ViT-B; MobileSAM vit_t (TDV masks); Walk-around canvas CNN (TorchPredic...

M	module	the check	window · how it decides	looks at (General Model)	follows (trackers)	pixels	own models
M18	safety huddle	Safety huddle conducted at huddle cone	before arrival · judges the window before the moment	cones, belt loader, GSE, pushback, trailer, fuel truck	aircraft, people	yes	none
M19	pre-departure walk-around	Pre-departure walk around completed	before departure · needs the whole event	belt loader, GSE, pushback, trailer, engine, cones, tail, nose	aircraft, people, belt loader	yes	YOLOv8m-pose (worker keypoints, sin...; YOLOv8m-pose rescue re-track (secon...; Depth Anything V2 metric ViT-B; MobileSAM vit_t (TDV masks); Walk-around canvas CNN best_acc.ker...
M20	pushback waits for wing walkers	Pushback does not start until wing walkers are in place a...	pushback / departure · verdict at the event	tail, wing, pushback, front wheel, belt loader, GSE, trailer, o...	aircraft, people	no	none
M21	pushback pathway clear	Pushback pathway confirmed clear of obstacles	pushback / departure · one detection → Fail	pushback, tail, nose, belt loader, GSE, fuel truck, trailer, co...	aircraft	no	none
M22	handrails fully extended	Safety handrails fully extended and used	belt loader at the door · verdict at the event	people, trailer, GSE, rear wheel	aircraft, belt loader	yes	EfficientNetB0 handrail-extension c...
M23	safety vests	Employees wearing safety vests secured to body	whole session · one detection → Fail	vests	people	yes	Swin-T orientation classifier (back...; Swin-T vest classifier (unzipped / ...; ResNet-18 vest-presence gate; YOLOv8m-pose (v4 belly-visible gate); ResNet-18 small-band vest CNN (v4_c...
M24	safety zone clear	Safety zone confirmed clear	before arrival · judges the window before the moment	cones, front wheel, people, pushback, belt loader, GSE, fuel tr...	aircraft	yes	MobileSAM vit_t (SamPredictor, box ...
M26	steering bypass pin	Steering by-pass pin installed, or steering otherwise byp...	arrival · Fail if not seen in time	people, front wheel, chocks	aircraft, people	yes	HRNet-W48 top-down heatmap pose (CO...; tsai InceptionTime pin-installation...; MinMaxScaler
M27	wing walkers with wands	Wing walkers in proper position and using approved wands	pushback / departure · judges the window before the moment	engine, wing, nose, tail, pushback, belt loader, GSE, trailer, ...	aircraft, people	no	HRNet-W48 top-down heatmap pose (CO...

4 · How long each check runs, and whether it already runs live

Post job: one clean job on the 36-minute reference event. Batch: median cost per recorded frame when the monthly test set runs (the usual way they run today). Live: the module's own work per frame inside the real-time branch, the shared models excluded. Runs live as is: whether the unchanged module gives the same verdict in real time.

M	module	post job	batch per frame	live per frame	runs live as is
M23	safety vests	370 s	28.2 ms	18.8 ms	yes, but only answers when the camera stops
M14	handrails on GSE used	249 s	30.7 ms	12.5 ms	yes, but only answers when the camera stops
M27	wing walkers with wands	111 s	3.2 ms	5.1 ms	yes
M13	hand signals	91 s	10.1 ms	0.3 ms	no — reads the recording twice
M24	safety zone clear	83 s	10.3 ms	0.2 ms	no — reads the recording twice
M12	GSE parked and chocked	76 s	6.1 ms	3.2 ms	yes, but only answers when the camera stops
M15	lead marshaller and wing walkers	59 s	5.7 ms	2.5 ms	yes
M22	handrails fully extended	46 s	8.4 ms	2.4 ms	yes
M02A	aircraft chocks (3 verdicts)	38 s	5.7 ms	0.5 ms	no — reads the recording twice
M07	conditioned air removed	27 s	2.4 ms	1.3 ms	yes

M	module	post job	batch per frame	live per frame	runs live as is
M16	pushback pin verification	27 s	3.9 ms	0.4 ms	no — reads the recording twice
M26	steering bypass pin	26 s	1.9 ms	0.7 ms	yes
M18	safety huddle	22 s	1.9 ms	0.9 ms	yes
M11	FOD walk	21 s	3.0 ms	1.3 ms	yes
M08	cones removed after GSE clear	20 s	4.9 ms	1.5 ms	yes
M01	3-stop brake check	16 s	3.4 ms	0.6 ms	yes
M21	pushback pathway clear	16 s	3.4 ms	0.5 ms	yes (the report shifts slightly)
M09	cones placed	13 s	2.3 ms	0.6 ms	yes (the report shifts slightly)
M20	pushback waits for wing walkers	13 s	2.4 ms	0.3 ms	yes
M10	crew present	12 s	1.2 ms	0.5 ms	yes
M04	belt loader forward chock	12 s	2.7 ms	0.3 ms	yes
M05	belt loader rear cone	12 s	—	0.4 ms	yes
M03	cargo doors opened	8 s	1.4 ms	0.9 ms	yes
M06	chocks and cones staged	7 s	1.4 ms	0.1 ms	yes
M17	post-arrival walk-around	own environment	6.8 ms	—	no — its own Python environment
M19	pre-departure walk-around	own environment	26.1 ms	—	no — its own Python environment

Where the time goes. Two modules dominate: safety vests (6 min on the reference event — two transformer classifiers on every person of every frame, the whole session) and handrails on GSE (4 min — four networks per belt-loader visit). The twelve checks that read only detections cost seconds each. In batch the busiest part of a turnaround is the belt loader at the door: nine checks active together, 89 ms of work per recorded frame against a real-time budget of 125 ms. The shared General Model and trackers are about 85 % of a real-time frame; the modules' own work is the small part.

5 • The models inside the modules

Thirteen modules carry models of their own, on top of the shared layer. Size is the weights file.

M	module	model	kind	version / date	MB
M02A	aircraft chocks (3 verdicts)	frame-difference chock classifier (rear wheels)	image classifier	—	—
M07	conditioned air removed	AC hose detector (YOLO11s-seg, 1 class)	segmentation	—	—
M11	FOD walk	MoveNet SinglePose Thunder v4	pose (body keypoints)	—	25
M12	GSE parked and chocked	frame-difference chock classifier	image classifier	—	—
M12	GSE parked and chocked	RTMPose-l body pose (17 keypoints)	pose (body keypoints)	—	—
M13	hand signals	HRNet-W48 top-down heatmap pose (COCO-17)	pose (body keypoints)	2022-09-13	269
M14	handrails on GSE used	EfficientNetB0 handrail-extension classifier	image classifier	2023-09-26	—
M14	handrails on GSE used	SegFormer-style handrail segmentation (smp Unet, MiT-...	segmentation	2023-10-30	—
M14	handrails on GSE used	DWPose-l whole-body pose (133 keypoints incl. hands)	pose (body keypoints)	—	—
M14	handrails on GSE used	Hand-contact classifier (blind checks)	image classifier	—	—
M15	lead marshaller and wing walkers	HRNet-W48 top-down heatmap pose (COCO-17)	pose (body keypoints)	2022-09-13	269
M15	lead marshaller and wing walkers	worker role classifier (SVC)	image classifier	—	—

M	module	model	kind	version / date	MB
M15	lead marshaller and wing walkers	walk-along classifier (RandomForest)	image classifier	—	2
M15	lead marshaller and wing walkers	walker movement-start models (left, right)	model	—	4
M16	pushback pin verification	SimCC ResNet-50 top-down pose (COCO-17)	pose (body keypoints)	2022-09-13	178
M17	post-arrival walk-around	YOLOv8m-pose (worker keypoints; singleton shared with...	pose (body keypoints)	—	—
M17	post-arrival walk-around	Depth Anything V2 metric ViT-B	depth estimation	—	—
M17	post-arrival walk-around	MobileSAM vit_t (TDV masks)	segmentation	—	—
M17	post-arrival walk-around	Walk-around canvas CNN (TorchPredictor CNNModel)	image classifier	—	—
M19	pre-departure walk-around	YOLOv8m-pose (worker keypoints, singleton)	pose (body keypoints)	—	—
M19	pre-departure walk-around	YOLOv8m-pose rescue re-track (second YOLO instance, B...	pose (body keypoints)	—	—
M19	pre-departure walk-around	Depth Anything V2 metric ViT-B	depth estimation	—	—
M19	pre-departure walk-around	MobileSAM vit_t (TDV masks)	segmentation	—	—
M19	pre-departure walk-around	Walk-around canvas CNN best_acc.keras via torch twin	image classifier	—	—
M22	handrails fully extended	EfficientNetB0 handrail-extension classifier	image classifier	2023-09-26	—
M23	safety vests	Swin-T orientation classifier (back / head / side)	image classifier	—	110
M23	safety vests	Swin-T vest classifier (unzipped / zipped)	image classifier	—	110
M23	safety vests	ResNet-18 vest-presence gate	image classifier	—	45
M23	safety vests	YOLOv8m-pose (v4 belly-visible gate)	pose (body keypoints)	—	53
M23	safety vests	ResNet-18 small-band vest CNN (v4_cnn2)	image classifier	—	45
M24	safety zone clear	MobileSAM vit_t (SamPredictor, box prompt)	segmentation	—	—
M26	steering bypass pin	HRNet-W48 top-down heatmap pose (COCO-17)	pose (body keypoints)	2022-09-13	269
M26	steering bypass pin	tsai InceptionTime pin-installation classifier	image classifier	—	2
M26	steering bypass pin	MinMaxScaler	time-series classifier	—	—
M27	wing walkers with wands	HRNet-W48 top-down heatmap pose (COCO-17)	pose (body keypoints)	2022-09-13	269

Shared and repeated. One pose network (HRNet-W48, 269 MB) is the same file in four modules — hand signals, steering, lead marshaller, wing walkers — each loading its own copy. EfficientNet-B0 appears six times for different jobs. MobileSAM is in four places. Five different pose stacks serve ten modules. Several weights files sit in the checkouts and are loaded by nobody (a second 255 MB HRNet in all four pose modules, a wand detector, a worker-type classifier, an air-unit detector).

6 • Where the same logic is written several times

23 of the 27 modules work out the stage of the turnaround themselves (when the aircraft arrived, when it left) instead of taking it from one place, and **19 reach into the tracker's internal fields** to do it — so every change of the tracker reaches them. The same arrival block is copied word for word in nine modules; six more have their own rules; the code base has five different definitions of “departure”. “Pushback attached” is computed twice, and that is why those two modules read the recording twice. “Belt loader at the door” exists in five variants. Two modules recompute the aircraft type although the General Model publishes it. The modules are pinned to ten different revisions of the shared library, and seven of the checkouts the monthly test runs differ from the revisions the review inspected.

M	module	what it recomputes on its own
M01	3-stop brake check	belt loader at the door
M02A	aircraft chocks (3 verdicts)	arrival, pushback attached

M	module	what it recomputes on its own
M03	cargo doors opened	arrival, departure
M04	belt loader forward chock	belt loader at the door, aircraft type
M05	belt loader rear cone	belt loader at the door
M06	chocks and cones staged	arrival
M09	cones placed	arrival, departure
M10	crew present	arrival
M11	FOD walk	arrival
M12	GSE parked and chocked	departure, GSE stop / departure
M13	hand signals	arrival, belt loader at the door, aircraft type
M14	handrails on GSE used	belt loader at the door
M15	lead marshaller and wing walkers	arrival, departure
M16	pushback pin verification	pushback attached
M17	post-arrival walk-around	arrival, departure, belt loader at the door, its own tracking
M18	safety huddle	arrival
M19	pre-departure walk-around	arrival, departure, belt loader at the door, its own tracking
M20	pushback waits for wing walkers	arrival, departure, pushback attached
M21	pushback pathway clear	departure
M22	handrails fully extended	belt loader at the door
M24	safety zone clear	arrival, its own tracking
M26	steering bypass pin	arrival
M27	wing walkers with wands	arrival, departure

7 · What the review flagged: what the code verifies against what the checklist says

M	module	note
M01	3-stop brake check	Do not replace its recomputed status with generic tracker status without replay comparison; seconds vary with configured FPS.
M02A	aircraft chocks (3 verdicts)	Placement is assessed within 30 seconds of aircraft stop.
M03	cargo doors opened	Does not verify a BL target-door relationship or independently distinguish mechanical open/closed state beyond detector-label semantics.
M04	belt loader forward chock	A one-time chock latch does not prove continuous chocking throughout the entire parked interval.
M05	belt loader rear cone	Checklist and code thresholds differ; preserve code behavior as the migration baseline, then approve requirement changes separately.
M06	chocks and cones staged	YAML 6/4 counts are not the active main-code 7/6 counts.
M07	conditioned air removed	Held boxes are counted repeatedly in current history.
M08	cones removed after GSE clear	Despite its name it does NOT test whether GSE is chocked, and its equipment coverage is narrower than all GSE.
M09	cones placed	Obstacle report text says 40% while code tests 65%; inconsistent return arity requires an explicit legacy adapter.
M10	crew present	This measures counts over a window, not continuous presence of the same three named workers for ten minutes.
M11	FOD walk	Not the simple horizontal-crossing rule in the basic description; bbox-derived scale is not calibrated physical distance.
M12	GSE parked and chocked	Different chock evidence channels have different semantics; current truthy fusion must be made explicit before sharing.

M	module	note
M13	hand signals	Raised wrists are a coarse gesture proxy, not an approved signal vocabulary.
M14	handrails on GSE used	Whole-body hands are required; do not substitute body-only pose.
M15	lead marshaller and wing walkers	No separate pose/wand model or lead-marshaller signal proof on the active path.
M16	pushback pin verification	Measures a verification-like worker action, not direct pin visibility.
M17	post-arrival walk-around	Post-arrival rescue/outlier options are disabled by default.
M18	safety huddle	Pinned cv_common ac5098d lacks imported PlaneArrivalDetector.
M19	pre-departure walk-around	Pinned cv_common e4c5e36 lacks imported departure/presence helpers.
M20	pushback waits for wing walkers	Current evidence is post-start co-motion, not a direct proof that everyone was ready before start.
M21	pushback pathway clear	Current hazard class list does not include people.
M22	handrails fully extended	480 frames is about 60 seconds at 8 FPS, not eight minutes.
M23	safety vests	No keypoint pose despite the model name.
M24	safety zone clear	Upstream tracker does not cover every transport class currently tracked here.
M26	steering bypass pin	Distinct learned installation task from heuristic pin verification.
M27	wing walkers with wands	CRITICAL semantic gap: approved-wand Pass is hard-coded on the active path.